Inkomoko Early Warning System (ML) Progress Presentation

Capstone Team

Semester 4 Simulations

February 19, 2026

Agenda

1. Project Overview
2. Current Repository Status
3. Modeling Progress
4. Data and Documentation
5. Quality, Gaps, and Next Steps
6. Current Model Performance
7. Appendix

Project Goal

Objective

Build a machine learning early-warning pipeline to monitor client business health and predict short-term risk and impact outcomes.

Core ML Tracks

- Revenue prediction
- Employment prediction
- Risk scoring and tiering

Expected Value

- Earlier intervention signals
- Better portfolio monitoring
- Data-driven support prioritization

Repository Snapshot (Attached ML Folder)

Top-level scripts	`dummy_data_generation.py`, `requirements.txt`
Model outputs	`artifacts/` with joblib models + metrics + predictions CSVs
Experimentation	`notebooks/` for revenue, employment, risk, and RAG agent
Documentation	`Docs/EMPLOYMENT DOC.md`, `REVENUE DOC.md`, `RISK DOC.md`
Presentation assets	`prezi/prezi.tex`
Synthetic data	`synthetic_outputs/core_banking_loans.csv`, `impact_data.csv`

What Has Been Completed

- Synthetic data generation pipeline exists.
- Separate notebooks created for each ML task:
  - `revenue-pred-model.ipynb`
  - `employment-pred-model.ipynb`
  - `risk-model.ipynb`
- Trained model artifacts exported to `artifacts/`.
- Test predictions exported as CSV files.
- Model metrics and summary files exported.
- Domain documentation completed for all three tracks.

Revenue Modeling Progress

Evidence of Progress

- Notebook: `notebooks/revenue-pred-model.ipynb`
- Metrics: `artifacts/revenue_model_metrics.csv`
- Predictions: `artifacts/revenue_predictions_test.csv`

Current status:

- Baseline modeling and evaluation pipeline established.
- Artifact export workflow is functional.
- Ready for feature tuning and validation hardening.

Employment Modeling Progress

Evidence of Progress

- Notebook: `notebooks/employment-pred-model.ipynb`
- Models:
  - `employment_jobs_created_3m_model.joblib`
  - `employment_jobs_lost_3m_model.joblib`
  - `jobs_created_3m_model.joblib`
  - `jobs_lost_3m_model.joblib`
- Metrics + predictions exported to CSV.

Risk Modeling Progress

Evidence of Progress

- Notebook: `notebooks/risk-model.ipynb`
- Models:
  - `risk_score_3m_model.joblib`
  - `risk_tier_model.joblib`
- Portfolio-level output:
  - `segment_summary.csv`
  - `model_summary_metrics.csv`
  - `predictions_test.csv`

Data Assets Used

Synthetic Outputs

- `core_banking_loans.csv`
- `impact_data.csv`

Generation Script

- `dummy_data_generation.py`

Purpose

- Enable modeling without production dependency
- Support repeatable testing
- Facilitate fast prototyping of features

Documentation Coverage

Document	Coverage
`EMPLOYMENT DOC.md`	Employment target definitions, assumptions, and interpretation guide
`REVENUE DOC.md`	Revenue prediction framing, features, and output usage
`RISK DOC.md`	Risk score/tier design, segmentation, and warning interpretation

Progress Indicator

All three core model streams have documentation artifacts present and versioned.

Current Strengths and Gaps

Strengths

- End-to-end ML flow exists
- Multi-target modeling implemented
- Saved models + predictions + metrics
- Track-specific documentation available

Gaps

- Centralized training pipeline/script not yet explicit
- Cross-model dashboarding not yet integrated
- Threshold calibration and drift checks pending
- Deployment packaging not yet shown in repo

Planned Milestones (Next Sprint)

1. Consolidate notebooks into reusable training/inference scripts.
2. Standardize feature preprocessing and model versioning.
3. Compare model families and tune top performers.
4. Define early-warning thresholds and intervention rules.
5. Build one reporting layer (CSV-to-dashboard/API-ready output).
6. Add validation checks: data quality, leakage guardrails, drift baseline.

Requested Decisions / Support

- Confirm success criteria per track (revenue, employment, risk).
- Approve early-warning threshold framework.
- Confirm preferred deployment target (batch pipeline vs API).
- Validate priority user stories for intervention workflows.

Current Model Performance

Revenue

Missing:
`revenue_model_performance.png`

Employment

Missing:
`employment_model_performance.png`

Risk

Missing:
`risk_model_performance.png`

Attached File Inventory (As Provided)

```
dummy_data_generation.py
requirements.txt
.ipynb_checkpoints/
    requirements-checkpoint.txt
artifacts/
    employment_jobs_created_3m_model.joblib
    employment_jobs_lost_3m_model.joblib
    employment_model_metrics.csv
    employment_predictions_test.csv
    jobs_created_3m_model.joblib
    jobs_lost_3m_model.joblib
    model_summary_metrics.csv
    predictions_test.csv
    revenue_model_metrics.csv
    revenue_predictions_test.csv
    risk_score_3m_model.joblib
    risk_tier_model.joblib
    segment_summary.csv
charts/
Docs/
    EMPLOYMENT DOC.md
    REVENUE DOC.md
    RISK DOC.md
notebooks/
    employment-pred-model.ipynb
    rag-agent.ipynb
    revenue-pred-model.ipynb
    risk-model.ipynb
    .ipynb_checkpoints/
        employment-pred-model-checkpoint.ipynb
```

Thank you

Questions and feedback.